

Mapping the Convergence of Organizational Strategies and AI: A Bibliometric Analysis

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Abstract—As AI continues to reshape organizational decision-making and strategic planning, understanding its research landscape becomes increasingly vital. This study offers a bibliometric analysis of global research at the intersection of Artificial Intelligence (AI) and organizational strategy. Using tools Bibliometric analysis with Biblioshiny by R studio, it maps thematic clusters, author contributions, and international collaborations. AI and Machine Learning emerge as dominant themes, with growing interest in Generative AI and sustainability. The U.S., India, and China lead in publication output, while collaboration networks show fragmented but expanding global engagement. Findings highlight AI's central role in strategic innovation and decision-making, advocating for inclusive research partnerships and strategic investment in global knowledge ecosystems.

Keywords—Artificial Intelligence, Organizational Strategy, Bibliometric Analysis, Machine Learning, Generative AI, Strategic Decision-Making

I. INTRODUCTION

Companies see the use of Artificial Intelligence (AI) as a key part of their plans. This change altered the way people talk about management today. Businesses face unstable environments filled with data. AI enables us to explore new possibilities. They improve predictive analytics, automate decisions and also enhance strategy flexibility. Traditional strategic frameworks are changing. Examples include SWOT analysis, Porter's Five Forces, and the Resource-Based View. AI-driven insights enhance scenario planning [1]. They allocate resources effectively and strengthen their position in the market. This collaboration marks a change as we shift from gut feelings to strategic plans based on facts.

Recent studies show that more people are interested in this area. Recent bibliometric analyses show an increase in publications that explore AI's impact on organizational processes, leadership, and innovation ecosystems [2]. Researchers identified key themes related to digital transformation, machine learning applications, and strategic governance. This field consists of individuals from various disciplines. Bibliometric tools such as Biblioshiny in R Studio illustrate the connections between ideas and the networks of collaborators. They give a big picture view of how AI changes how people think strategically in different fields [3].

AI has an effect on how well things work. It also affects how organizations are designed and how well they can plan for the future. McKinsey's strategic framework says that AI is an important part of improving every step of strategy development. This encompasses environmental scanning, implementation, and performance evaluation [4]. AI systems work as researchers, interpreters, and simulators. They help people think better. Companies can feel more sure about making bold strategic moves. Leaders need to change the way they do things as things change. Companies also need to improve their skills. This will help them make the most of AI's power.

This chapter explores global research trends. This study focuses on how AI aligns with organizational strategy. This study looks at how often publications appear. It highlights the importance of key authors. It also tracks how themes change over time. This study maps the intellectual landscape of this convergence. The results help us learn more about AI. It changes the way str

ategic tools work. It alters our strategic thinking in the digital era. This study offers valuable insights for academics, professionals, and policymakers. They aim to ensure that new technologies align with long-term and adaptable business plans.

II. REVIEW OF LITERATURE

Researchers focus on strategically integrating Artificial Intelligence (AI) into organizations. This subject includes business, technology, and policy. Recent research highlights how AI contributes to developing dynamic capabilities. It prepares businesses for market changes. They can quickly shift resources as needed [5]. Previously, people relied on their instincts for strategic responsiveness. Now, it connects more to simulations and predictive modelling based on data. These changes indicate a conflict between new digital thinking and traditional strategic reasoning. Researchers study organizational adaptability. They recognize AI as a key factor in transforming various sectors. These sectors include healthcare, retail, and manufacturing. Zhuang et al. [6] present evidence indicating that the implementation of AI substantially influences operational decision-making. It promotes decentralized strategy implementation and strengthens organizational resilience. Their results show a change. Algorithms are now used in management models. Strategic decisions arise from intricate data ecosystems. They do not arise from hierarchical deliberation.

Research highlights the significance of AI in enhancing supply chain strategies. Stakeholders become more involved. Zangiacomi et al. [7] examined the impact of AI-driven analytics on supply chain coordination. Researchers found that these tools simplify risk management and enhance flexibility. AI relies on its ability to sense and respond. Many people view it as a method to differentiate themselves from others. Many people debate the moral implications of these changes. This holds particularly when discussing data privacy. It is crucial to ensure algorithms are accountable. Recent studies indicate that researchers increasingly explore the intersection of AI and strategic management. Vidwans et al. [8] conducted a detailed analysis of citations and keywords in publications from 2012 to 2024. The researchers observed changes in the themes. These included AI governance, strategic foresight, and organizational learning. Their research indicates that the field consists of individuals from various disciplines. This process depends heavily

on real-time decision feedback loops. It is important to be strategically flexible. It stresses how important it is to keep track of things over time. This method helps us figure out how ideas grow and spread in the mind. These ideas will help guide future research. They also make sure that technological advances are in line with the creation of organizational strategies. Discussions on AI-driven strategic transformation highlight the significance of dynamic capabilities. It discusses organizational models that prioritize new ideas. Alhyasat et al. [9] performed a systematic review of the literature. They found three main groups of research. These groups are interested in how AI can give companies an edge over their competitors, how new technologies affect strategic management, and how dynamic capabilities work. The results demonstrate that AI enhances strategic foresight and flexibility. Appio et al. [10] examined AI-powered models that manage research, development, and engineering. They demonstrated the impact of predictive analytics and real-time monitoring on strategic planning and resource allocation. Ranaweera [11] examined the influence of AI on the strategy of the semiconductor industry. The research underscores AI's capacity to transform manufacturing and design methodologies. Ranade and Saravia [12] performed a systematic review regarding the instruction of AI ethics in professional communication. They stressed how important it is to be ethically literate when using AI. These studies show how AI can change a lot of things about how a business works. It has an effect on optimizing operations and running things in an ethical way.

III. RESEARCH QUESTIONS

- To visualize and understand the conceptual structure of this research landscape through co-occurrence network.
- What are the prevailing thematic clusters within the research landscape.
- Who are the most relevant and important authors in this research domain?
- What are the trend topics that are prevalent over a period of time in this research landscape?
- Which countries have collaborated and are important in this study.
- Evaluate the connection between authors, countries and keywords in this domain.

IV. RESULTS AND INTERPRETATION

The first objective is to visualize and understand the conceptual structure of this research landscape through co-occurrence network.

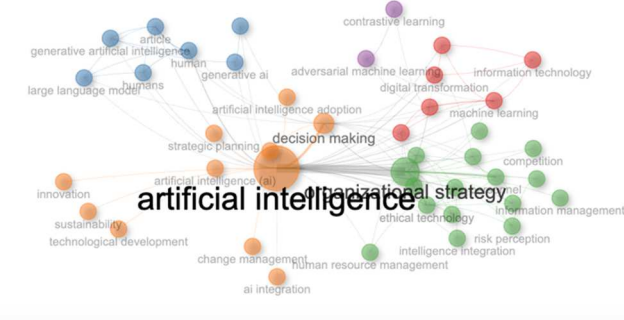


Fig. 1. Co-occurrence Network.

The keyword co-occurrence network shows how often two or more terms show up in the same publication. It shows how the research landscape is set up in terms of ideas. The size of the node shows how often keywords show up. The thickness of the edge shows how strong their co-occurrence is. This is normalized to show important connections. The central term that holds the network together is "Artificial Intelligence." The walk trap clustering algorithm finds groups of people who are interested in the same things. The orange group is all about AI. It talks about things like making decisions, coming up with new ideas, being environmentally friendly, and adoption. A big part of the green cluster is managing information. It talks about morals, how people view risk, and how businesses change. The red group is all about technical details. It has things like machine learning, IT, and digital change. The blue cluster has generative AI, big language models, and words that are about people. This means that more and more people are interested in new AI tools. There is both adversarial and contrastive learning in a small purple cluster. These parts highlight specific aspects of AI. These parts draw attention to certain parts of AI. Overall, the network underscores AI's central role and its integration into organizational strategy, innovation, and technical development.

The next section evaluates the prevailing thematic clusters within the research landscape through the map and also see the motor themes, niche themes, basic themes and emerging or declining themes.

The thematic map sorts keyword clusters into four quadrants based on two axes: centrality (x-axis) and development (y-axis). The four quadrants are motor themes (central and well-developed),

basic themes (central but less developed), niche themes (developed but less central), and emerging/declining themes (low centrality and development). Artificial Intelligence and Machine Learning are two of the most important themes, which means they are both important and well-developed.

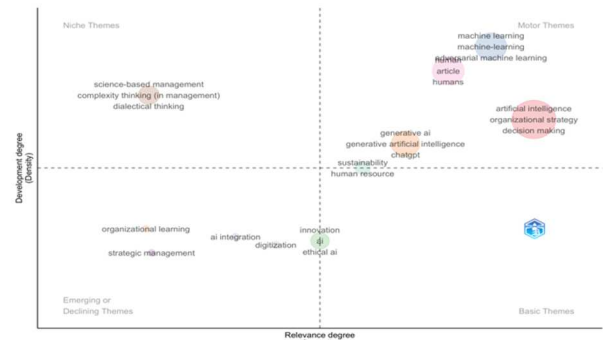


Fig. 2. Thematic Map.

AI includes terms like organizational strategy and decision-making. Key articles by Lourens, Ma sri, and Surie are important. Machine Learning includes adversarial machine learning and human-centered parts, with help from Alotaibi, Liashenko, and Jashapara. The human theme, which is closely related to AI and machine learning, is based on research in ethics and healthcare, with help from Garcia, Cross, and Chen. Generative AI is a new trend. It becomes more important as it grows. Cui, Agrawal, and Biel write about how it affects businesses. Mollah's work emphasizes sustainability as a central theme in AI research. Innovation and digitization are themes that are either growing or fading. People are starting to pay more attention to newer terms like "digital transformation." The bottom left quadrant is where organizational learning and strategic management are. This placement shows that these areas are not very important. Science-based management is a field of study. It is well-developed, but it doesn't have a lot of power. Laske's study exemplifies this domain. The analysis utilizes the top 250 keywords derived from the SCOPUS dataset. It doesn't include terms that are used less often and treats variations separately. The walk trap algorithm finds groups of things. Layout parameters make sure that things are clear and that nodes don't overlap.

The next section evaluates the most relevant authors and their production overtime in this research landscape.

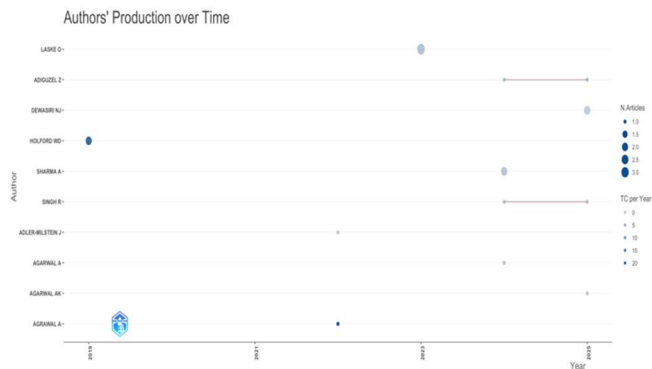


Fig. 3. Relevant Authors

The author's production timeline reveals a rapidly expanding research field. We see significant growth in artificial intelligence, digital transformation, and related technologies. Numerous publications emerged in the years following 2019. In 2024 and 2025, we see a significant increase. This pattern indicates increased interest and higher scholarly output in recent years. Scopus has limited historical data before 2019. This suggests that the field may be relatively new. Changes in indexing practices may occur. Authors might shift their focus toward emerging topics.

The timeline allows us to distinguish between established researchers and newcomers. Authors with longer red lines and a consistent publication history are likely more established contributors. Authors with shorter lines and recent entries may be newcomers to the field. This distinction clarifies leadership dynamics. It also highlights influence and knowledge sharing in the research community. Examining each author individually clarifies the situation further. Agrawal A's article "Digital Twin: From Concept to Practice" appears in the Journal of Management in Engineering. It was published in 2022. This article has the highest number of citations, with a TCpY of 22.5.

This author published three articles in 2019. This shows ongoing engagement. Agrawal AK has one publication in 2025. This indicates a recent entry into the field. The article titled "The Future of Work: Generative AI's Influence on Turnover Intentions and Organizational Strategy" has not accumulated citations yet. This is expected due to its recent publication. Adler-Milstein J published one paper in 2022. It has not received any citations so far. The topic centers on health system digitization. It may be niche or less visible in broader research circles. Agarwal A published one article in 2024. The article focuses on AI and digital transformation. It has a moderate

TCpY of 0.5. Co-authored works indicate collaborative research efforts. Singh R demonstrates activity in 2024 and 2025. However, the low citation numbers hinder an assessment of the impact of their contributions.

Sharma A co-authored the same article as Agrawal A. This suggests a duplicated entry or shared authorship. Holford WD has one of the earliest publications in the dataset from 2019, with a relatively high TCpY of 17.6. This work appears in Futures. It examines the digital economy and shows significant impact. Dewasiri NJ publishes in 2025. They focus on AI adoption, governance, and reskilling in at least two papers. Recent publications likely explain the low citation counts. Adiguzel Z published works in 2024 and 2025. These works focus on AI, big data, and sustainable design. The citation impact is still limited. Laske O has a group of publications from 2023. However, there is little citation activity associated with them. Their emphasis on advanced systems-level problem solving and dialectical thinking indicates a theoretical or methodological approach. This may attract a more specialized academic audience. The next section delves into the trend topic and how they have evolved over time.



Fig. 4. Trend Topics.

The plot shows the most popular topics from the SCOPUS dataset in 2024. It uses the 'KW_Merged' field to highlight keywords with the highest annual median frequency. The size of the bubble shows how important a keyword is. Lines show how different publications can be different from each other. "Artificial Intelligence" stands out as a key term. It shows that research will be very important in 2024. Kwik-E-Mart. Anonymous-journalist. prof, AudienceB. !! 1. implications of ples.... apolog-someone. namedboards. announcement. nominations estim apology Legion Glas announcementNAP implet. 2 pouch 2 komodo 2 please There are a fair number of keywords, such as "Decision Making" and "Organizational Strategy." This shows that there is a significant, but not overwhelming, interest in these areas. When the bubbles are smaller, words like "Article" and "Personnel" don't mean much. AI is playing a bigger

r and bigger role in making decisions and planning. This trend shows that people are becoming more interested in how AI will affect organizational and strategic processes. The main point of the story is the “KW_Merged” field in SCOPUS. It shows clearly what the most important research areas are. It also talks about new trends that affect more than one field.

The next section will evaluate the country collaboration in this research landscape.

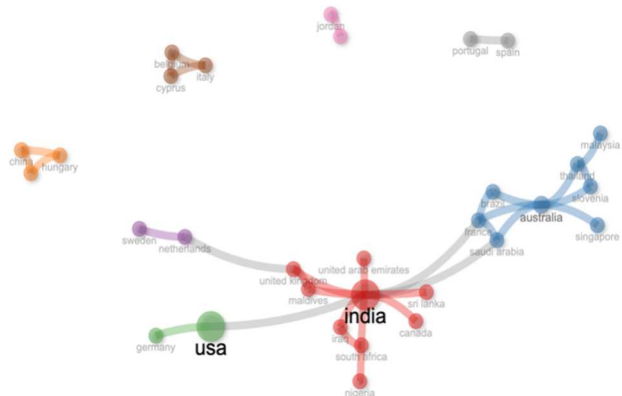


Fig. 5. Collaboration Network (Countries)

The international collaboration network is not very well organized. A lot of clusters don't connect. Research partnerships often include certain groups of countries. They don't have a strong global network. This fragmentation is caused by geography, historical ties, shared interests, and targeted funding. Countries exhibit varying degrees of centrality. India and Australia serve as key hubs.

They work with a lot of different countries. They facilitate broader research exchange. Countries like Jordan, China, and Hungary are on the outside. This implies they engage in limited or specialized collaborations.

The walk trap clustering algorithm finds groups of people who are different from each other. It does this by looking at how people work together. India is at the center of the red cluster. The UAE, UK, Maldives, Sri Lanka, Canada, Iraq, South Africa, and Nigeria are all part of it. These countries have strong ties, probably because of their history or because they share a theme. Australia, Brazil, France, Saudi Arabia, Malaysia, Thailand, Slovenia, and Singapore are all in the blue group. This setup makes it clear that Australia is a bridge between continents. The green cluster shows that the US and Germany have a strong partnership. It does a good job of showing the relationship between the two countries. There are smaller groups, like China-Hungary (orange), Sweden-Netherlands (purple), Belgium-Italy-Cyprus (brown), and Portugal-Spain (grey). These likely represent regional or niche collaborations. It looks like Jordan is alone. This means that there isn't much international involvement.

India and Australia emerge as key players in their respective clusters, while the USA, though mainly linked to Germany here, likely maintains broader global research ties beyond this network.

The next section delves into a three D plot showing relation between some authors, countries and keywords.

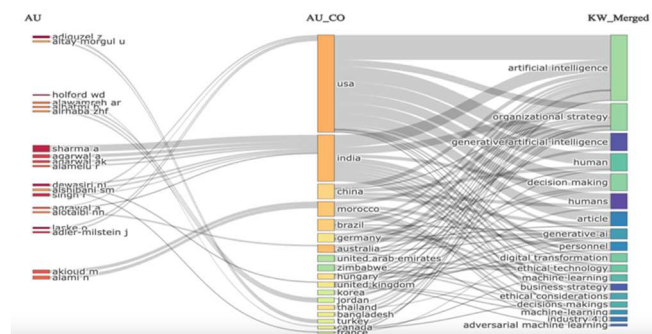


Fig. 6. Three D Plot

The visualization includes three main fields: authors, countries, and keywords. These fields are connected by ribbons that show how they are related. The lengths and heights of the bars show how many times a keyword appears in a publication, and the thickness of the ribbons shows how strong the connections are. The United States is the best at publishing. India and China are not far behind. “Artificial Intelligence” is the most common term, which shows how important it is in the dataset. Other significant terms include “Organizational Strategy” and “generative artificial intelligence.” Strong author-country links help American researchers make important contributions. Country connections show what themes are important. AI and strategy are linked to the US. India is linked to generative AI. There may not be a direct link between authors and keywords. But you can guess what their research topics are based on the countries they are from.

V. CONCLUSION

The bibliometric visualizations show a dynamic and interconnected research landscape. This landscape exists at the intersection of Artificial Intelligence and organizational strategy. Authors, countries, and keywords show that the United States leads in contributions. India and China follow closely behind. Artificial Intelligence stands out as a key term. It plays a central role in strategic discussions. Meanwhile, Organizational Strategy and Generative AI are gaining importance. Links between authors and countries show c

oncentrated expertise. American researchers stand out in this regard. Thematic priorities differ by region. For example, India emphasizes generative AI as a significant trend. The global collaboration map highlights important research hubs in North America, Europe, and East Asia. It shows increasing participation from India, Southeast Asia, and South America. The U.S., Europe, and China maintain strong ties. Regional initiatives such as Horizon 2020 promote robust international partnerships. Africa has limited connectivity. However, there are clear efforts to promote inclusion and build capacity. The findings support fair and strategic investment in global research ecosystems. They promote inclusive growth and encourage diverse regions to contribute meaningfully to the future of AI-driven organizational strategy.

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